

# MAT 426, Class 1 Homework: Solving Systems with LU Factorization

(Due Tuesday, February 10, 2026)

In Exercises 1–3, solve the system  $A\mathbf{x} = \mathbf{b}$  using the given factorization of  $A$ .

1.  $A = \begin{bmatrix} 4 & -2 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 4 & -2 \\ 0 & 4 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} 0 \\ 8 \end{bmatrix}$

2.  $A = \begin{bmatrix} 2 & -4 & 0 \\ 3 & -1 & 4 \\ -1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ -\frac{1}{2} & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & -4 & 0 \\ 0 & 5 & 4 \\ 0 & 0 & 2 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ -5 \end{bmatrix}$

3.  $A = \begin{bmatrix} 2 & -1 & 0 & 0 \\ 6 & -4 & 5 & -3 \\ 8 & -4 & 1 & 0 \\ 4 & -1 & 0 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 4 & 0 & 1 & 0 \\ 2 & -1 & 5 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & -1 & 5 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}; \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$

In Exercises 4–5, use the process in Example 3, Class 1 handout to find an LU factorization of  $A$ .

4.  $A = \begin{bmatrix} 2 & 5 \\ -3 & -4 \end{bmatrix}$

5.  $A = \begin{bmatrix} 3 & -1 & 2 \\ 3 & -2 & 10 \\ 9 & -5 & 6 \end{bmatrix}$

6. Let  $A$  be an  $n \times n$  lower triangular matrix with nonzero entries on the diagonal. Show that  $A$  is invertible and that  $A^{-1}$  is also lower triangular.
7. Let  $A$  and  $B$  be  $n \times n$  lower triangular matrices. Show that  $A \times B$  is also an  $n \times n$  lower triangular matrix.